

Dated: _____

QUESTION No. 5.

Pseudocode

```
CAP A = 20, CAP B = 40, CAP C = 15
DECLARE N, VT, UC, Permit, SEMI: CHARACTER
DECLARE isValid, CP, allocated, FULL: BOOLEAN
OCC A = 0, OCC B = 0, OCC C = 0, C = 0, B = 0, V = 0, A = 0,
rej = 0, Total = 0
DECLARE remCap, i, remA, remC, maxOCC, INTEGER
DECLARE assignedZone, highest: STRING
Print ("Enter total expected vehicles")
INPUT N

FOR i = 1 TO N DO
    Total = Total + 1
    REPEAT
        Print ("Enter Vehicle type (C for Car, B for Bike,
            V for Van)")
        INPUT VT
        Print ("Enter User Category (F for Family, S
            for student, G for Visitor)")
        INPUT UC
        Print ("Has Valid Permit (Y/N):")
        INPUT Permit

        isValid = (VT = "C" OR VT = "B" OR VT = "V") AND
            (UC = "F" OR UC = "S" OR UC = "G")
            AND (Permit = "Y" OR Permit = "N")

        IF
            NOT isValid
        THEN
```

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```
Print("Invalid Input. Please reenter")  
END IF  
UNTIL invalid.
```

```
CP = FALSE // can proceed.
```

```
IF Permit == "Y"
```

```
THEN
```

```
CP = TRUE
```

```
ELSE
```

```
Print("Is it an Emergency Vehicle(Y/N):")
```

```
INPUT Em
```

```
// Emergency
```

```
IF Em == "Y" THEN
```

```
THEN
```

```
CP = TRUE
```

```
ELSE
```

```
Print("Vehicle rejected")
```

```
Rej = Rej + 1
```

```
ENDIF
```

```
ENDIF
```

```
IF CP
```

```
THEN
```

```
allocated = FALSE
```

```
assigned zone = ""
```

```
IF UC == "T"
```

```
THEN
```

```
IF (Cap A - OccA) >= 1
```

```
THEN
```

```
assigned zone = "ZoneA"
```

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allocated = TRUE
END IF

ELSE IF VC == "S"

THEN

IF VT == "V"

THEN

IF (CapC - OccC) >= 2

THEN

OccC = OccC + 2

assigned Zone = "Zone C"

allocated = TRUE

ENDIF

ELSE

IF (CapB - OccB) >= 1

THEN

OccB = OccB + 1

assigned Zone = "Zone B"

allocated = TRUE

ENDIF

ENDIF

ELSE IF

VC == "G" THEN

IF VT == "V" THEN

IF (CapC - OccC) >= 2 THEN

OccC = OccC + 2

assigned Zone = "Zone C"

allocated = TRUE

ENDIF

ELSE

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```
IF (CapC - OccC) >= 1 THEN  
    OccC = OccC + 1  
    assignedZone = "Zone C"  
    allocated = TRUE  
ENDIF  
ENDIF  
ENDIF
```

```
IF allowed THEN  
    Acc = Acc + 1  
    IF VT == "C" THEN  
        Cars = Cars + 1  
    ELSE IF VT == "B" THEN  
        Bikes = Bikes + 1  
    ELSE  
        Vans = Vans + 1  
    ENDIF  
ENDIF  
ENDIF
```

```
IF assignedZone == "Zone A" THEN  
    rem = CapA - OccA  
ELSE IF assignedZone == "Zone B" THEN  
    rem = CapB - OccB  
ELSE  
    rem = CapC - OccC  
ENDIF  
ENDIF
```

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Print ("Assigned:" assigned Zone, "Remaining Capa-
-city:", rem)

ELSE

Print- ("Vehicle rejected")

rej = rej + 1

ENDSE

ENDSE

ENDFOR

remA = CapA - OccA

remB = CapB - OccB

remC = CapC - OccC

highest = "ZoneA"

max Occ = OccA

IF OccB > maxOcc

THEN

maxOcc = OccB

highest = "ZoneB"

ENDIF

IF OccC > maxOccC

Then

maxOcc = OccC

highest = "ZoneC"

ENDIF

IS FULL = (remA == 0 AND remB == 0 AND remC == 0)

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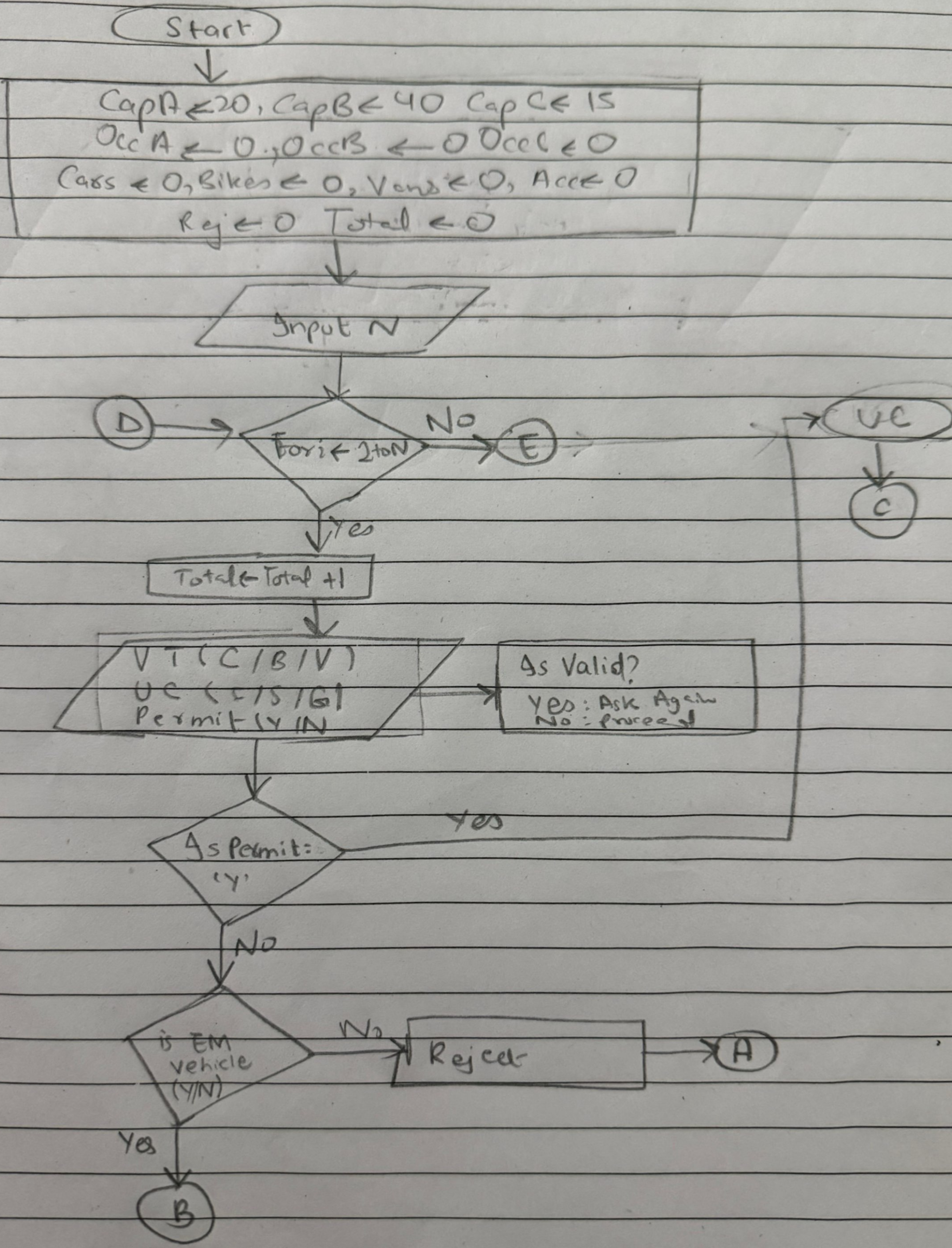
```
Print ("Total", Total)
Print ("Accepted" Acc)
Print ("Rejected" rej)
Print ("Cars" Cars "Bikes" Bikes "Vans" Vans)
Print ("Zone A occupancy:" OccA "remaining" remA)
Print ("Zone B occupancy:" OccB "remaining" remB)
Print ("Zone C occupancy:" OccC "remaining" remC)
Print ("Zone with highest occupancy", highest).
```

IF is FULL THEN

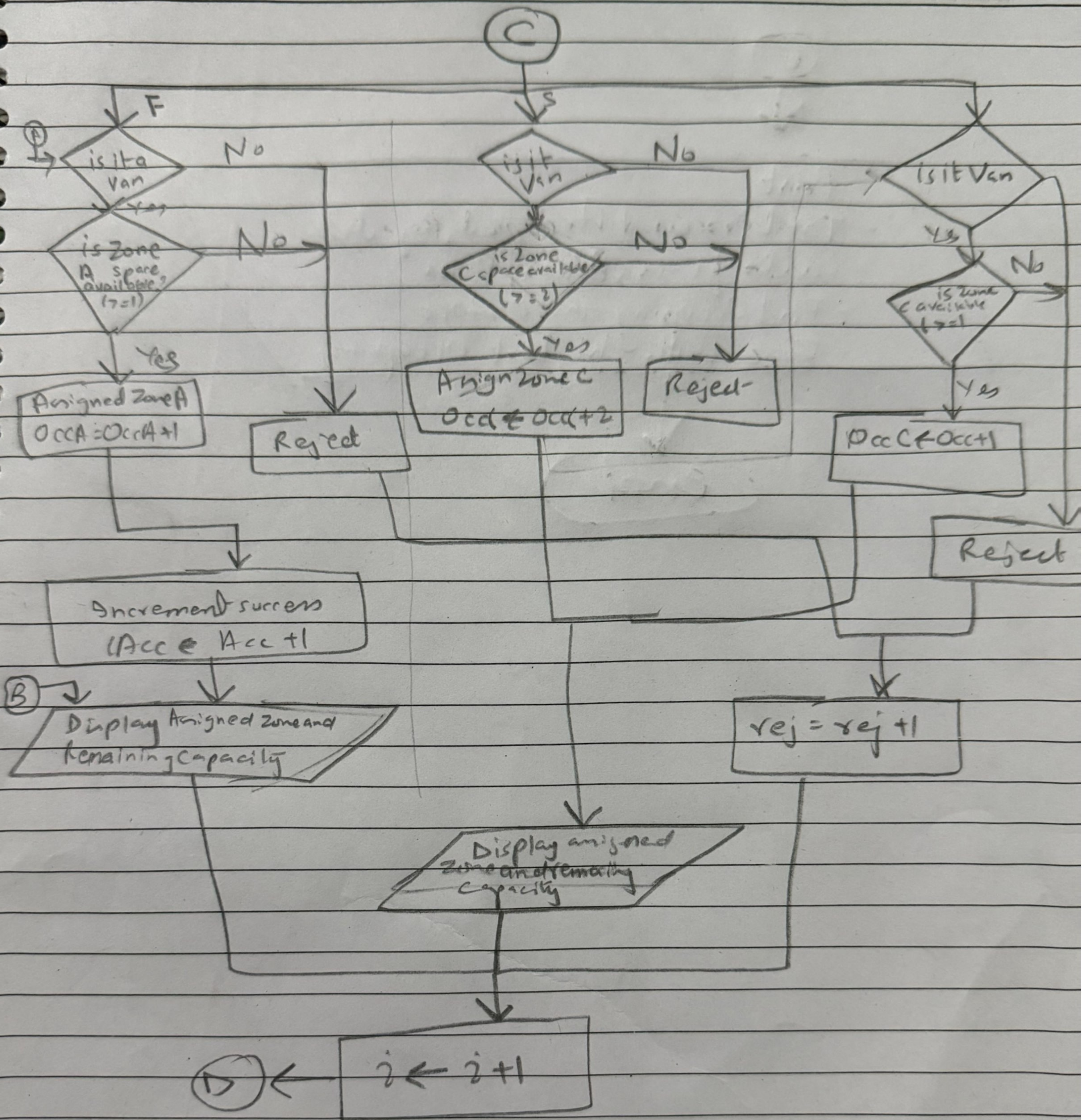
Print ("campus Facility status : FULL")

ELSE

Print ("Campus Facility status : Available").



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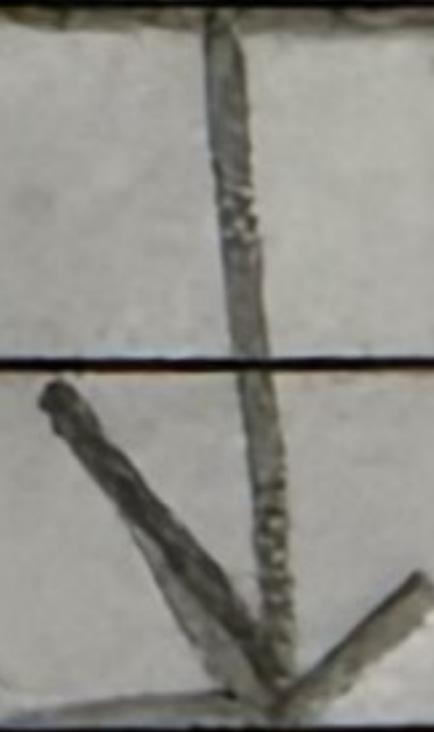
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⑤



Print

- Total Processed, Accepted, Rejected
- Counts of Cars, Bikes, Vans
- Find Occupancy and Remove
- Zone With highest Occupancy



end

INPUT	OUT PUT
* Total expected vehicles: N * Vehicle Type: VT (C for car, B for Bike, V for Van) * User Category: UC (F for faculty, S for student, G for Visitor) * Permit status: Permit (Y/N) * Emergency status: Em (Y/N), asked if Permit = "N" * fixed capacities: Cap A = 20, Cap B = 40, Cap C = 15	* Validation message (eg, error prompts) * Per-vehicle status: Assigned zone and remaining capacity, or rejection notification * Total processed vehicles (Total) * Total accepted vehicles (Acc) and total rejected vehicles (rej) * Accepted counts by type (Cars, Bikes, Vans). * final occupancy and remaining capacity for Zones A, B, and C. * Zone with highest occupancy (highest) * Overall campus facility status (FULL or Available)

PROCESSING REQUIRED	SOLUTIONS ALTERNATIVE
* TO * Input validation: Repeat input until $VT \in \{C, B, V\}$ AND $UC \in \{F, S, G\}$ AND Permit $\in \{Y, N\}$ * Eligibility check: IF Permit == "Y" or Em == "Y", set CP = TRUE;	* Loop structure: A Counted loop (for i = 1 To N) to iterate through all incoming vehicles. * Validation structure: A REPEAT... UNTIL post-test loop to filter invalid

otherwise reject vehicle
(re)

* Zone assignment rules

(if $CP == "F"$): Assign

Zone A if space available

($CapA - OccA \geq 1$); update

occupancy.

- Student

($UC == "S"$): If $VT == "V"$,

assign Zone C (needs ≥ 2

spaces, $OccC$); else assign

Zone B (needs ≥ 1 space,

$OccB$).

- Visitor

($UC == "G"$): If $VT == "V"$,

assign Zone C (needs ≥ 2

spaces); else assign Zone C

(needs ≥ 1 space)

* Update accepted vehicle

counters: Ac , and increment

vehicle category counters

(Cars, Bikes, Vans).

* Compute remaining capacities

$rem = Cap - Occ$ for each zone.

* Determine maximum occupancy

Compare $OccA, OccB, OccC$ to

Character inputs:

* Capacity checking:

Checking threshold limits

before incrementing counters

to prevent parking overflows.

* Multi-space allocations: Vans

require 2 slots instead of 1

when parking in Zone C.

Set highest.

* facility fullness check:

is1,

INPUT	PROCESS REQUIRED	OUTPUT
* N (Total vehicles)	1) Initialize counters:	* Prompt messages.
* VT (Vehicle type)	To; Ac, re, Ca, Bil, Va', and	(validation errors,
* UC (User category)	zone occupancies Oc', Oc', Oc',	input requests)
* Permit (Has permit: Y/N)	2) Read N.	* Vehicle allocation or rejection messages.
* Preset Capacities: CapA=20, CapB=40, CapC=15.	3) Loop N times (FOI): a, Read and validate VT, Uc, Permit. b) Evaluate eligibility: Check permit status; if absent, query and evaluate emergency status (Em). Set CP flag. c) If eligible (CP=TRUE): - Match user category and vehicle type against zone allocation constraints. - Verify zone availability (≥ 1 spot for cars/bikes, ≥ 2 spots for vans in zone c). - If allocated: Update specific zone occupancy, mark a?, increment vehicle	* Assigned zone and remaining space per vehicle * Summary metrics: - Total vehicles processed (Total) - Total accepted (Acc) - Total rejected (rej) - Vehicle breakdowns (Cars, Bike, Vans) - Pre-zone occupancy and remaining slots

type counter (Cars, Bikes, or Vans), and increment Acc. (Zone A, Zone B, Zone C)

- If not allocated: Increment rej. - Zone with highest occupancy (highest)

d) If ineligible: Increment rej. - Overall facility

e) Compute remaining capacity status (Full or for the assigned zone and Available)

display immediate assignment or rejection.

4) Calculate final remaining Capacities: remA, remB, remC;

5) Determine the zone with the maximum occupancy (highest)

6) Determine facility status (is1).

7) Display aggregated summary reports.